

Homework 11

Nov 26, 2023

1 (30'). Define the problem QUADEQ as follows: given m quadratic equations in n variables over the finite field $\text{GF}(2)$,

$$\begin{aligned} p_1(x_1, x_2, \dots, x_n) &= 0, \\ p_2(x_1, x_2, \dots, x_n) &= 0, \\ &\vdots \\ p_m(x_1, x_2, \dots, x_n) &= 0, \end{aligned}$$

determine whether there exists at least one solution. Prove that QUADEQ $\in$ NP.

2 (30'). Prove if L is NP-hard and $L \in \text{P}$, then $\text{P} = \text{NP}$.

3 (40'). Prove that $L_1 \leq_P L_2$ and $L_2 \in \text{NP}$, then $L_1 \in \text{NP}$.